



Original Article

Prediction of nitrofurantoin resistance among *Enterobacteriaceae* and mutational landscape of *in vitro* selected resistant *Escherichia coli*



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ABSTRACT

Nitrofurantoin (NIT) has long been a drug of choice in the treatment of lower urinary tract infections. Recent emergence of NIT resistant *Enterobacteriaceae* is a global concern. An ordinal logistic regression model based on PCR amplification patterns of five genes associated with NIT resistance (*nfsA*, *nfsB*, *ribE*, *oqxA*, and *oqxB*) among 100 clinical *Enterobacteriaceae* suggested that a combination of *oqxB*, *nfsB*, *ribE*, and *oqxA* is ideal for NIT resistance prediction. In addition, four *Escherichia coli* NIT-resistant mutants were *in vitro* generated by exposing an NIT-susceptible *E. coli* to varying concentrations of NIT. The *in vitro* selected NIT resistant mutants (NI2, NI3, NI4 and NI5) were found to have mutations resulting in frameshifts, premature/lost stop codons or failed amplification of *nfsA* and/or *nfsB* genes. The *in vitro* selected NI5 and the transductant colonies with reconstructed NI5 genotype exhibited reduced fitness compared to their parent strain NS30, while growth of a resistant clinical isolate (NR42) was found to be unaffected in the absence of NIT. These results emphasize the importance of strict adherence to prescribed antibiotic treatment regimens and dosage duration. If left unchecked, these resistant bacteria may thrive at sub-therapeutic concentrations of NIT and spread in the community.

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1. Introduction

Nitrofurantoin (NIT), which was originally introduced as a broad spectrum antibiotic in the 1950s to treat uncomplicated lower urinary tract infections (UTIs), has been reintroduced to combat antibiotic resistant infections [1]. At therapeutic concentrations of 200 mg/L, NIT acutely inhibits the growth and multiplication of even the resistant strains [2]. Previous reports indicate that clinically significant resistance against NIT was relatively uncommon [3,4]. More recently, emergence of NIT resistance among clinically isolated bacterial pathogens are being reported [5]. A recent report indicated that resistance to NIT is an indicator of the underlying extensively drug resistant (XDR) phenotype in *Enterobacteriaceae* [6]. The role of several molecular determinants in the emergence of NIT resistance has been previously described [3,7–9]. However, a

comprehensive analysis that simultaneously delineates the contribution of all the reported NIT resistance genes assumes utmost importance to gain further insights into this phenomenon and would assist in the prediction of NIT resistance among *Enterobacteriaceae*.

Antibacterial activity of NIT is due to its conversion into lethal intermediates by the bacterial oxygen insensitive nitroreductases NfsA and NfsB [8]. NfsA (26.8 kDa) and NfsB (24 kDa) are flavin mononucleotide (FMN)-containing enzymes that use NADP/NADPH as electron donors. The nitro anion free-radical and hydroxylamine intermediates generated upon the processing of NIT were reported to interfere in replication, transcription, translation and carbohydrate metabolism leading to the eventual bacterial cell death [10].

NIT resistance is developed by the acquisition of mutations in *nfsA* and *nfsB* genes [8]. These mutations help the enzymes avoid processing of the nitrofurans class of drugs and reduce the accumulation of toxic intermediates inside the bacterium. Mutations in the *nfsA* and/or *nfsB* genes among *Escherichia coli* isolates either by

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insertion or substitutions were found to confer resistance against NIT due to the synthesis of inactive nitroreductases [7]. Additionally, a 12-nucleotide deletion in *ribE* gene encoding lumazine synthase involved in the biosynthesis of flavin mononucleotide (FMN) was detected among *in vitro* induced NIT resistant isolates [3]. Plasmid-mediated *oqxAB* encoding a multidrug efflux pump across different species of bacteria, possibly acquired through horizontal gene transfer were found to enhance NIT resistance [9]. The combination of *oqxAB* genes and *nfsAB* mutations were found to confer robust NIT resistance [1,9,11] (Fig. 1).

In this study, the combinatorial effect of *nfsA*, *nfsB*, *ribE*, *oqxA* and *oqxB* on NIT resistance has been investigated among 100 clinical *Enterobacteriaceae* isolates and the predictive potential of these genes has been assessed. Further, *in vitro* selected resistant mutants were screened for mutations in their *nfsA*, *nfsB* and *ribE* genes. Also, the fitness of *in vitro* selected mutants and P1 phage transductants at varying concentrations of NIT has been analysed.

2. Materials and methods

2.1. Bacterial strains

Based on the nitrofurantoin resistance and/or susceptibility, 100 *Enterobacteriaceae* [34 NIT sensitive (NS), 21 NIT intermediate resistant (NIR) and 45 NIT resistant (NR)] (Supplementary Table 1) were obtained from the collection of clinical isolates previously described [6]. These isolates were used to develop a PCR based NIT resistance prediction model. Biochemical identification and antibiotic susceptibility (GN-280 AST cards) of all the isolates were performed using Vitek 2.0. These isolates belonged to six different species, i.e., *E. coli*, *Klebsiella pneumoniae*, *Klebsiella oxytoca*,

Enterobacter cloacae, *Enterobacter aerogenes* and *Citrobacter koseri*, exhibiting varying ranges of resistance against NIT (Table 1). *E. coli* ATCC 35218 was used as a control strain. All methods were carried out in accordance with the CLSI guidelines [12].

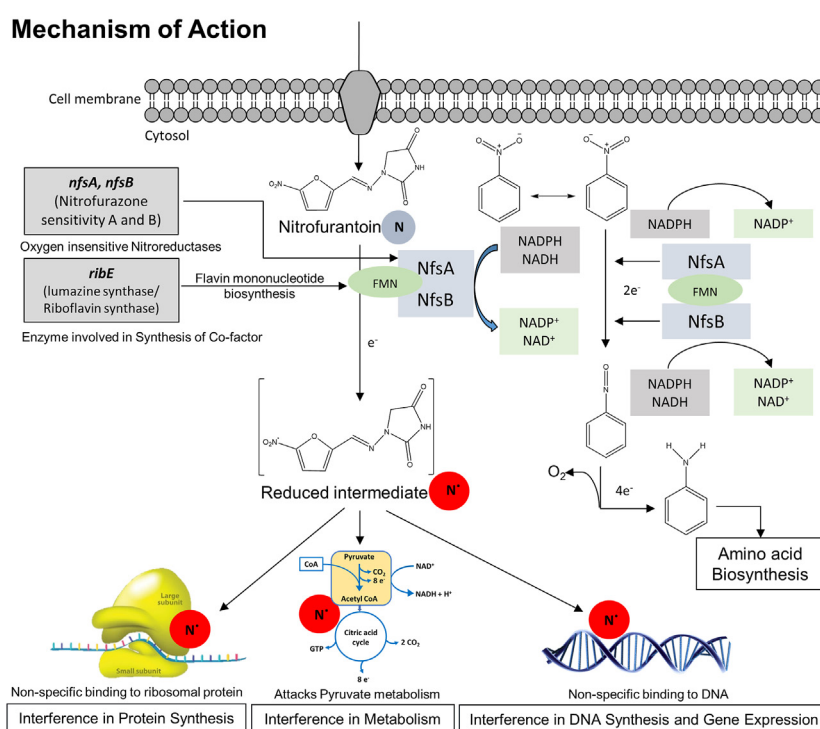
2.2. Extraction of genomic DNA and polymerase chain reaction

Genomic DNA was extracted from overnight cultures (3 ml) of all the 100 clinical isolates in LB broth using Macherey–Nagel NucleoSpin tissue DNA extraction kit as per the manufacturer's protocol. Genomic DNA from each of the isolates was used as template to amplify *nfsA*, *nfsB*, *ribE*, *oqxA* and *oqxB* genes (primers – Supplementary Table 2) and their amplification (standardised PCR conditions – Supplementary Table 3) results were noted as Y (positive amplification) or N (no amplification/size variation). Two independent sets of primers with different annealing sequences and PCR conditions were used in a step wise manner to obtain unambiguous results. First, all the isolates were screened with primer set 1. PCR amplification patterns of *oqxA* and *oqxB* genes using primer set 1 in the study isolates are taken from our previous

Table 1
Details of bacterial isolates analyzed in the study.

Isolate	Nitrofurantoin, MIC (mg/L)			Total
	Sensitive <16 to 32	Intermediate Resistant 32< to ≤64	Resistant >64	
<i>Escherichia coli</i>	32	11	12	55
<i>Klebsiella sp.</i>	1	7	28	36
<i>Enterobacter sp.</i>	1	1	4	6
<i>Citrobacter koseri</i>	0	2	1	3
Total	34	21	45	100

Mechanism of Action



Resistance

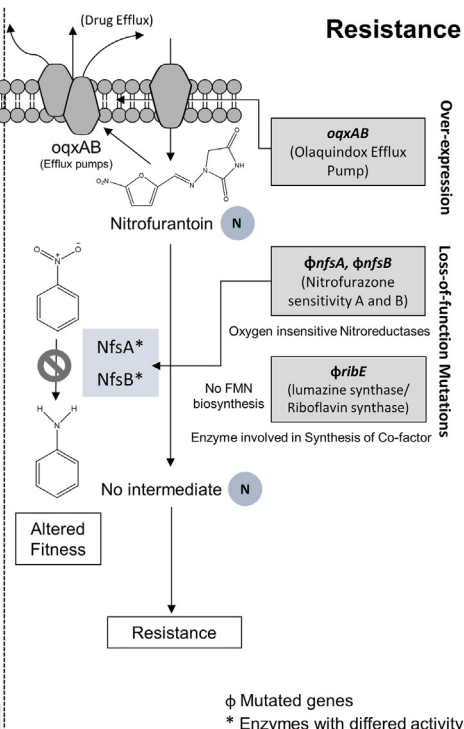


Fig. 1. Mode of action of nitrofurantoin and nitrofurantoin resistance mechanism. **Sensitivity:** Nitrofurantoin accumulated in the bacterial cytosol undergoes stepwise reduction by oxygen insensitive nitroreductases (NfsA and NfsB). The intermediates produced during these processes inhibit bacterial growth by affecting the DNA/protein synthesis or interfere in cellular metabolic pathways. RibE enzyme is involved in the synthesis of FMN, which acts as a cofactor for NfsA and NfsB. **Resistance:** Bacteria develop resistance against nitrofurantoin by acquisition of mutations in *nfsA*, *nfsB* and/or *ribE* to avoid processing of nitrofurantoin. Presence of OqxAB efflux pump further enhances the resistance by pumping the accumulated nitrofurantoin out of bacterial cell.

study [6]. PCRs, which resulted in amplification result 'N' using the primer set 1 for any of the five study genes, were repeated using the primer set 2. Amplification results that were found to be negative (N) with both the primer sets were considered to be truly negative and further analysis was done accordingly.

2.3. Statistical analysis

Pearson's chi-square test was used to determine association a) between the amplification result (Y or N) of all the five tested genes and NIT susceptibility phenotypes (NS, NIR and NR); and b) between each pair of genes. An ordinal logistic regression model was developed with all the five genes as variables for NIT resistance and it was assessed using step Akaike information criteria (AIC), with smaller values of AIC implying a better predictive combination of genes. At first, a constant was determined using the 'polr' function without any variables, followed by addition of each gene to the model and inspecting the AIC value to arrive at the combination of genes, which could provide the lowest AIC value, i.e., the best predictive model. All the analyses were performed in 'R'.

2.4. Selection of *in vitro* resistant mutants

Single and double step resistant mutants were selected by culturing NIT susceptible *E. coli* NS30 and *E. coli* ATCC 35218 on Mueller-Hinton agar containing a range of NIT (1, 2, 4, 6, 8, 16, 32, 64, 96, 128 and 256 mg/L) concentrations as described previously [13]. MICs of the *in vitro* generated resistant mutants were re-confirmed by Vitek 2.0. To account for growth medium bias, the parent NS30 was medium adapted in MH broth/agar without NIT for 10 passages (similar to the resistant mutants). To reconstruct the NI5 genotype in a clean background, P1 phage lysate of NI5 was used for transduction [14] of the parent NS30 and five independent NIT resistant transductant colonies (at 64 mg/L NIT) were selected.

2.5. Sanger sequencing for determination of mutations

Sanger sequencing (3130 XL Genetic Analyser, Thermo Fisher Scientific) of *nfsA*, *nfsB* and *ribE* amplicons together with their 150bp upstream and downstream regions were performed for NS30 *E. coli* and the *in vitro* selected resistant mutants through commercial vendors.

2.6. Growth curve analysis

To understand the effect of mutations causing NIT resistance, growth curve analysis was performed in 100 ml cultures (MH broth) with NIT sensitive *E. coli* ATCC 35218, parent and medium adapted NS30, NIT resistant NR42, *in vitro* selected resistant NI5 (MIC >512) and NS30 transductants (T1, T2, T3, T4 and T5) in a range of NIT concentrations (0, 64, 128, 200 mg/L). Absorbance (OD) was measured at 600 nm in a UV-Vis Spectrophotometer (SpectraMax M5) every 30 min from the time of inoculation until the culture reached stationary phase (8 h). On an hourly basis (0–8 h), 1 ml of the cultures were serially diluted 10^5 times and 100 μ l of the respective dilutions were plated on MH agar plates. The Colony Forming Units (CFUs) on each of these plates were counted after overnight incubation at 37 °C. To capture the entire series of events related to the growth curve, Area Under the Curve (AUC) was calculated for all the strains analysed for fitness at different concentrations of NIT. AUC was calculated using the formula $\Sigma (a+b/2 \times h)$ where $a = OD_{600}$ at time point 1, $b = OD_{600}$ at time point 2 and $h = 30$ min. All the readings were taken in quadruplicates.

3. Results

Positive amplification (Y) of *nfsA*, *nfsB* and *ribE* genes was assumed to reflect the susceptible phenotype of an isolate, while the resistant isolates were considered to possess variations (N) in the size of these genes. On the contrary, positive amplification (Y) of *oqxA* and *oqxB* genes was indicative of NIT resistant phenotype. The distribution of PCR amplification results (Y/N) among NS, NIR and NR isolates was analysed (Fig. 2) based on agarose gel electrophoresis results (Supplementary Table 4, Supplementary Figs. 1–2).

A strong association ($p < 0.01$) was observed between the resistance phenotype (NS, NIR and NR) and amplification patterns of *nfsA*, *nfsB*, *oqxA*, and *oqxB* genes (Supplementary Table 5). Analyses of pairwise association between the study genes revealed that the gene pairs *nfsA-nfsB*, *nfsA-ribE*, *nfsB-ribE* and *oqxA-oqxB* are strongly associated ($p < 0.01$) with each other (Supplementary Table 6), suggesting that variations in one of the two genes may not be independent of its partner gene. Further, in an ordinal logistic regression involving all the five study genes, status of *nfsB* ($p = 4.09E-03$) was found to significantly influence the outcome of NIT resistance (Supplementary Table 7). The model was assessed using Akaike information criteria (AIC), which estimates the in-sample prediction error and relative quality of statistical models for a given set of data. Sequential inclusion of *oqxB* (187.30), *nfsB* (172.32) and *ribE* (171.08) improved the model by lowering the AIC. The best AIC value (170.62) was achieved when *oqxA* was also added to this model, resulting in *oqxB-nfsB-ribE-oqxA* as the ideal combination of genes for the prediction of NIT resistance (NR). Although *nfsA* gene was found to have strong association (Supplementary Table 5) with NIT resistance independently, its exclusion from the model is justified due to strong association between *nfsA-nfsB* gene pairs ($p = 5.00E-04$). The accuracy of the model was found to be 91.1% (41 out of 45) for NIT resistant isolates and 91.2% (31 out of 34) for NIT sensitive isolates. However, 10 (47.6%) out of 21 NIT intermediate isolates were predicted as NIT resistant and the remaining 11 (52.4%) isolates were predicted as NIT susceptible.

Spontaneously acquired mutations in *nfsA*, *nfsB* and *ribE* genes were then investigated to understand the evolution of NIT resistance. Single step mutant (NI2) had an MIC of 32 mg/L and is considered susceptible to NIT, while the MIC of the double step mutants NI3 (256 mg/L), NI4 (>512 mg/L) and NI5 (>512 mg/L) suggested that they are highly resistant to NIT (Supplementary Fig. 3). The *nfsA*, *nfsB* and *ribE* gene sequences (Supplementary Table 8) of sensitive *E. coli* ATCC 35218 were used as reference for comparative sequence analyses. Several mutations were found to have accumulated in the *nfsA*, *nfsB* and *ribE* genes of the four *E. coli* resistant mutants (Table 2).

To evaluate the effect of acquired mutations on the fitness of *in vitro* selected mutants that are free from adaptive and compensatory mutations, genotype of NI5 was reconstructed in a clean background by transducing parent NS30 with P1 phage lysate of NI5. Five independent transductant colonies were selected on NIT (64 mg/L) containing MH agar media (Supplementary Fig. 4). No transductant colonies were observed at higher concentrations (128 mg/L) of NIT. Growth curves based on OD_{600} and CFUs were plotted (Fig. 3a–f) by culturing the bacteria across a range of NIT concentrations (0, 64, 128 and 200 mg/L). AUC based comparative growth curve analysis (Supplementary Table 9) revealed that *in vitro* selected resistant mutant NI5 had significant reduction in the growth rate compared to its parent NS30 (paired sample t-Test, $p = 2.75E-06$) and medium adapted NS30 (paired sample t-Test, $p = 2.46E-06$) strains. Similarly, remarkable reduction in the fitness was observed in the five transductants (AUC of T1-43, T2-30.8, T3-20.9, T4-47.3, and T5-83.8) when compared to the parent (AUC-

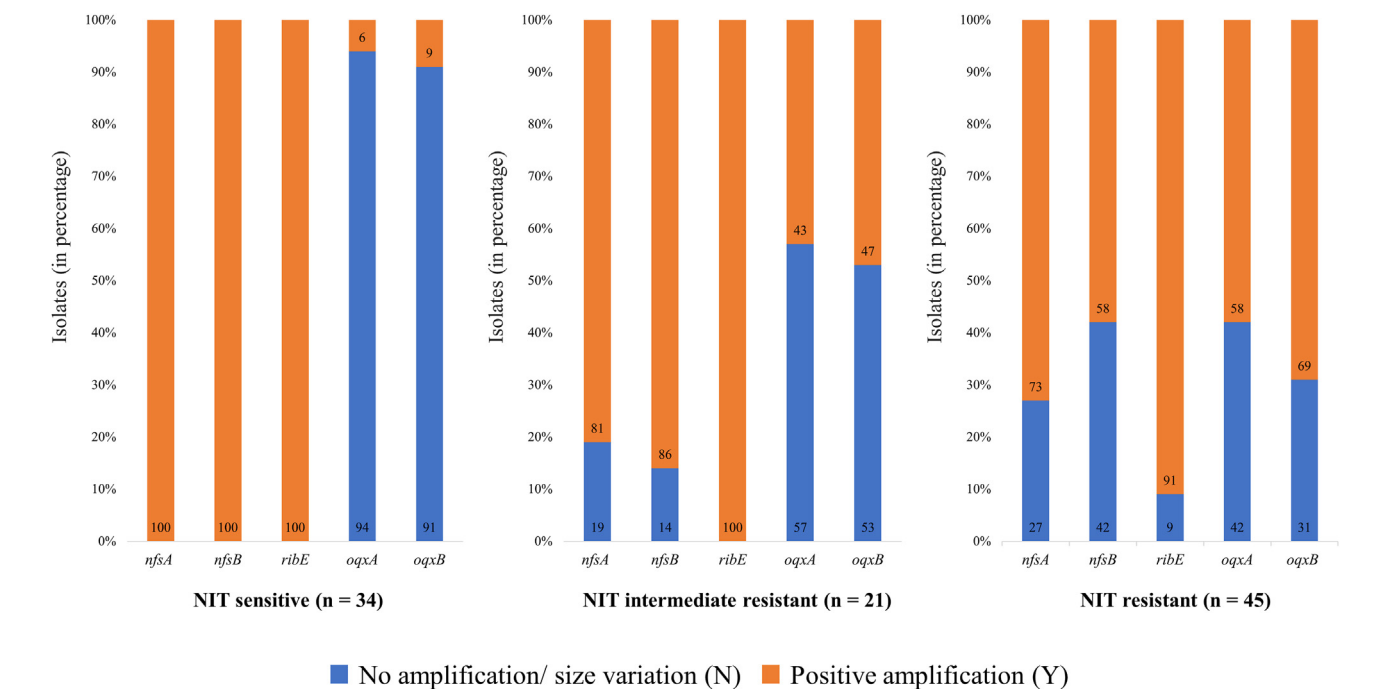


Fig. 2. Distribution of PCR amplification results (in percentage) of five study genes among NIT sensitive, intermediate resistant and NIT resistant isolates. PCR amplification of *nfsA* revealed that 27% of resistant (12 out of 45) and 19% of intermediate resistant (4 out of 21) isolates resulted in no amplification/size variation (N), while positive amplification (Y) was observed among all the NIT susceptible isolates (n = 34). Analyses for *nfsB* gene indicated that 42% of resistant strains (19 out of 45) and 14% of intermediate resistant strains (3 out of 21) had no amplification/size variation (N), while positive amplification (Y) of the gene was observed among all of the NIT susceptible isolates (n = 34). The *ribE* gene was found to be intact (Y) in all the NIT susceptible (n = 34) and intermediate resistant (n = 21) isolates. No amplification/size variation (N) of *ribE* was observed among 9% (4 out of 45) of NIT resistant isolates. Analyses for the presence or absence of OqxAB efflux pumps revealed that several of the resistant [58% – *oxxA* (26 out of 45); 69% – *oxxB* (31 out of 45)] and intermediate resistant [43% – *oxxA* (9 out of 21); 47% – *oxxB* (10 out of 21)] strains were found to possess *oxxA* and *oxxB* genes, while the same was not observed among NIT susceptible isolates [6% – *oxxA* (2 out of 34); 9% – *oxxB* (3 out of 34)].

Table 2
Comparison of *nfsA*, *nfsB* and *ribE* gene sequences of the select study strains with reference to *E. coli* ATCC 35218.

Strains	<i>nfsA</i>	<i>nfsB</i>	<i>ribE</i>
NS30 (Parent strain)	I117T, K141E, G187D, Insertion of CAATAAT at N ₄₀₈ 4 other synonymous SNPs	WT	N48D, Y107D, V117L
MIC <16 mg/L			
NI2 MIC = 32 mg/L	I117T, K141E, G187D, Insertion of CAATAAT at N ₄₀₈ 4 other synonymous SNPs	WT	N48D, Y107D, V117L
NI3 MIC = 256 mg/L	M1T 29 other missense mutations, 41 other SNPs, 1 deletion, insertion of CAATAAT	E175D, F176L, G177C, L178V, G182S, T184S, S185C, L186V, P190A, H193S, H194S, S195H, V196R, E197Stop	Y107D, V117L
	75 other insertions, Stop codon at N ₃₆₆ (122 aa truncated protein)		
NI4 (MIC >512 mg/L)	Partial sequence (411bp) obtained, 87% identity with <i>nfsA</i> of parent NS30 <i>E. coli</i>	P190L, V191L, S195I, N200R, A201V, T202R, L203S, P204V, L208H, P209R, Q210A, N211E, I212H, T213D, T215G, E216H, V217E, Stop218R	Y107D, V117L
NI5 (MIC >512 mg/L)	Failed Sequencing reaction	Failed Sequencing reaction	WT

WT: Wild Type; MIC: Minimum Inhibitory Concentration; SNP: Single nucleotide polymorphism.

124.85) and medium adapted NS30 (AUC-152.84) strains (Supplementary Fig. 5).

Importantly, there was no considerable impact on the fitness (AUC-644.3) of the resistant clinical isolate NR42 (MIC = 256 mg/L) when cultured in the absence of NIT. Its growth curve (Fig. 3b, Supplementary Table 9) was found to be similar (AUC-536.9) to that of susceptible laboratory strain *E. coli* ATCC 35218 (two sample t-Test, *p* = 0.23).

4. Discussion

The emergence of antimicrobial resistance (AMR) is a serious clinical challenge [15]. Development of lab-based AMR prediction tools which support clinicians towards early prescription of appropriate antibiotic therapy is urgently needed. In this regard, an attempt has been made to develop a simple, PCR based NIT resistance prediction model for *Enterobacteriaceae*. The proposed model

can be used in a clinical setting after due validation as it would help in reducing the time taken from sample collection to final diagnosis when compared to the conventional culture based identification and AST (Supplementary Fig. 6).

In one of the studies, NIT resistant clinical isolates were found to harbour deleterious mutations in *nfsA* gene alone or in combination with *nfsB* gene [7]. However, in another study by Osei Sekyere, 100% of the NIT resistant *Enterobacteriaceae* (31 nos) were found to have loss of function mutations in both *nfsA* and *nfsB* genes [1]. Among these, 18 isolates had mutations in the *ribE* gene [1]. In yet another analysis by Mottaghizadeh et al., six NIT resistant *E. coli* were found to have genetic alterations in the *nfsA* and *nfsB* genes, while alterations in the *ribE* gene was observed only in one of these isolates [16]. Notably, a number of NIT resistant clinical *Enterobacteriaceae* have been reported to harbour *oxxA* and *oxxB* efflux pumps [17–20].

All the reported genes (*nfsA*, *nfsB*, *ribE*, *oxxA* and *oxxB*) associated with NIT resistance [3,8,9] have been considered for this analysis.

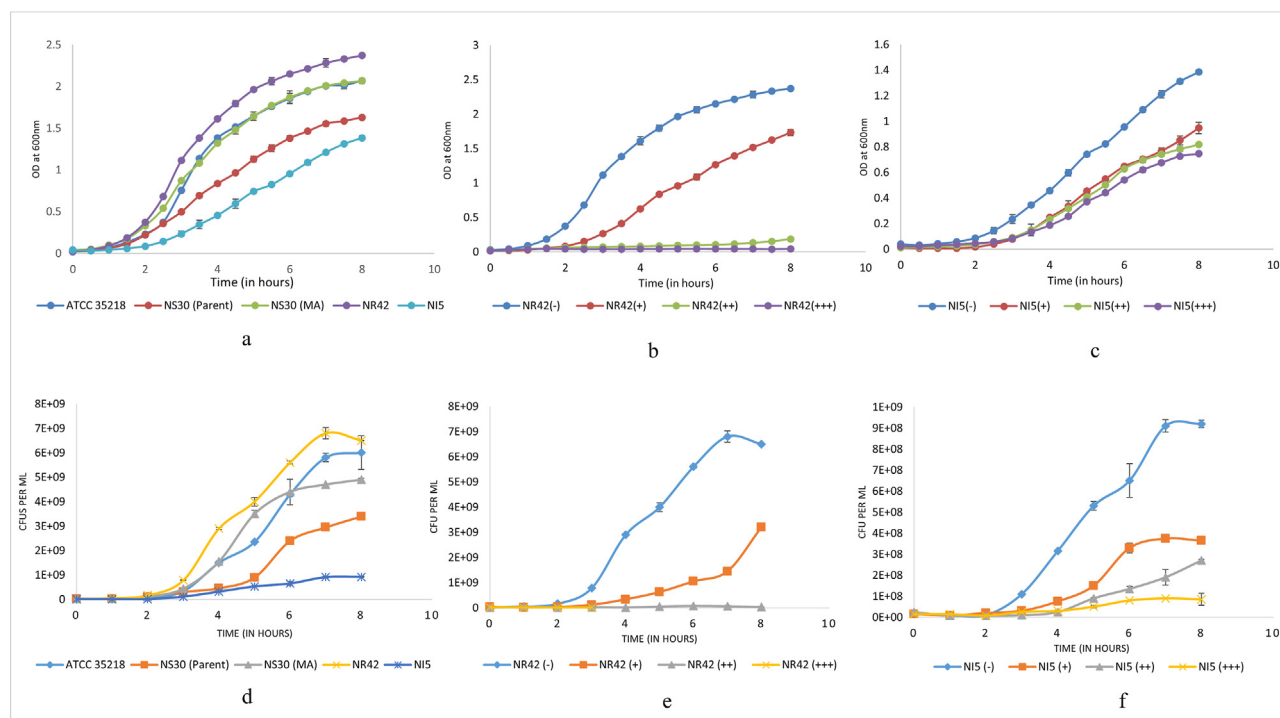


Fig. 3. Growth curves of ATCC 35218, NS30 (parent and medium-adapted), NR42 and NI5 *E. coli*. (–) MH without antibiotic (+) MH with 64 mg/L nitrofurantoin (++) MH with 128 mg/L nitrofurantoin (+++) MH with 200 mg/L nitrofurantoin. Growth curves of **a)** NIT sensitive *E. coli* ATCC 35218, parent and medium adapted (MA) NS30, *in vitro* generated NI5 mutant, and resistant clinical isolate NR42 in the absence of nitrofurantoin. NI5 was found to have a slower growth rate with prolonged lag phase compared to the other strains; **b)** resistant clinical isolate NR42 at different concentrations of NIT. Improved growth was observed when NIT was absent and at 64 mg/L compared to 128 mg/L and 200 mg/L of NIT; **c)** *in vitro* generated NIT resistant NI5 was cultured at different concentrations of NIT; prolonged lag phase was observed when NI5 was cultured in 128 mg/L and 200 mg/L of NIT, growth was found to improve in the absence of NIT; **d)** CFUs based growth curves of NIT sensitive *E. coli* ATCC 35218, parent and medium adapted (MA) NS30, *in vitro* generated NI5 mutant, and resistant clinical isolate NR42 in the absence of nitrofurantoin; **e)** CFUs based growth curves of resistant clinical isolate NR42 at different concentrations of NIT; **f)** CFUs based growth curves of *in vitro* generated NIT resistant NI5 at different concentrations of NIT.

Also, it was observed that NIT resistant bacteria had variations in the PCR amplicon size in *nfsA*, *nfsB* and *ribE* genes probably due to rearrangements (insertions/deletions/substitutions) [2,7,8]. The results from this study, in concordance with those reported previously [11,16], indicate that OqxAB efflux pumps have significant impact on NIT resistance and bacteria may have acquired these genes during the course of their evolution to overcome the crippling effects of NIT. PCR amplification based statistical analysis of 100 clinically isolated Enterobacteriaceae revealed the *oqxAB-nfsB-oqxA-ribE* combination as an ideal predictor of NIT resistant Enterobacteriaceae.

While resistant mutants of NIT sensitive clinical isolate NS30 were successfully selected *in vitro*, similar attempts using *E. coli* ATCC 35218 remained futile. Mutation frequency of NS30 was found to be $\sim 10^{-6}$ CFUs per generation, which is higher than previously reported studies [3], suggesting that the clinical isolates have an intrinsic flair for genetic rearrangements. Further investigations revealed that the NS30 *E. coli* had a seven-base insertion (CAATAAT) after T₄₀₅ in its *nfsA* sequence when compared to *E. coli* ATCC 35218. Although the 12-nucleotide deletion in *ribE* as reported previously [3] was not observed in any of our *in vitro* selected mutants, other point mutations were observed in the *ribE* gene of the resistant mutants. Taken together, the current study suggests that *nfsA* and *nfsB* are the first line of targets in the acquisition of spontaneous mutations and are critical in the emergence of resistance phenotype, while mutations in *ribE* are undertaken as a last resort by the bacteria when exposed to debilitating concentrations of NIT.

The current study sheds light on the intrinsic plasticity of genomes to perturb the existing architecture of specific genes in the process of responding to antibiotic stress among clinical isolates. Antibiotic resistance by the acquisition of mutations in the existing genes was found to be a stable trait in some cases [21], while in others it incurred fitness cost [2,3]. Acquisition of spontaneous mutations in *nfsA* and *nfsB* genes indeed affected the fitness of *in vitro* selected resistant strains. The resistant bacteria were found to survive albeit at a slower growth rate under therapeutic concentrations of NIT, but recovered when cultured without antibiotic (Fig. 3, Supplementary Table 9). Improved growth of the resistant clinical isolate (NR42) in the absence of NIT indicates that clinically resistant strains may have acquired ample compensatory mutations and adaptive strategies to thrive in the host environment. This emphasises the importance of strict compliance to the prescribed antibiotic treatment regimen and dosage duration to avoid the spread of resistant variants of previously antibiotic sensitive and treatable bacteria.

The proposed model is limited in segregating the resistant and susceptible variants with high accuracy (>91%), while the same is not true for intermediate resistant strains (47.6%). Further, the effect of SNPs and INDELs leading to minor variations in the size of the amplicons that are not visually detectable, or those that do not affect the primer annealing sites during amplification, are not accommodated. The model needs further validation using data from large number of isolates collected prospectively from multiple geographic locations before it can be used as a routine diagnostic tool in clinics.

Currently, diagnostic tests are designed to support treatment of hospitalized patients, but they are often not available in outpatient clinics [22]. Hence, PCR based rapid diagnostic tools which support the clinicians to shift from empirical to evidence-based treatments in outpatient clinics would help in the combat of AMR. Rapid ASTs were described to help in overall savings by reducing the hospital stay, morbidity and mortality [23]. Cost benefit analysis is necessary before rapid ASTs are introduced for routine clinical application among outpatient clinics. However, it is difficult to correctly estimate the total cost of these tests [22]. The current findings have significant clinical relevance in the context of antibiotic prescription. Remnants of the resistant mutants may easily disperse into the community and spread far and wide if appropriate dosage and compliance is not ensured; more specifically in the context of NIT resistance. Hence, great care is required while using these antibiotics. Strict adherence to prescribed protocols is of urgent need to extend the clinical lifetime of this very important antibiotic.

Ethical approval and informed consent

The “Institutional Ethics Committee of the Sri Sathya Sai Institute of Higher Learning” (ECR/616/Inst/AP/2014/RR-17) approved the study (SSSIHL/IEC/PSN/BS/2014/03) and also exempted the need for informed consent. The study is in accordance with the ethical standards of the Declaration of Helsinki.

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Transparency declarations

None.

Data availability

All data could be made available from the corresponding authors upon reasonable request.

Author contributions

This project was designed by B.E. Prevalence and S.A. Bacteria was isolated and cultured by BK and PK; antibiotic susceptibility test was done by BK and PPC; PCR screening was done by BK, SAG and SC; Resistance induction and growth curve experiments were done by BK and ASKP. Sequencing and analysis were done by ML and BK. Analysis of data, compilation, and manuscript was written by BK, SA and B.E.P. All authors approve the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resmic.2021.103889>.

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